

Executable Effect Summaries for Deterministic Parallelism

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Abstract

Static effect systems can guarantee deterministic parallelism, but region-level approximations often reject computations whose concrete run-time accesses are independent. Dynamic and optimistic systems recover more parallelism by observing concrete accesses, but typically require logging, validation, rollback, or replay. This paper explores a middle ground: *surface-effect terms*, source-level executable summaries that are statically checked against the computations they describe and evaluated before those computations are run.

We present a region-polymorphic simply typed λ -calculus with mutable references and executable effect summaries. The mechanized development proves that verified summaries approximate the dynamic traces of their paired computations, and that terminating continuation-machine executions are deterministic. Surface effects therefore provide a statically verified, source-level form of pre-execution access prediction: programmers may expose value-dependent access information before a computation is run, while the metatheory ensures that the summary covers the computation trace it claims to describe.

Contents

1	Motivating Example	4
2	Static Effects, Dynamic Traces, Surface Effects, and Computed Effects	5
3	Syntax	7
4	Operational Semantics	8
5	Static Semantics	11
6	Correctness of Surface Effects	12
7	Small-Step Runtime Safety	15
8	Discussion and Related Work	16
9	Conclusions	17
A	Deterministic Parallelism	18

Introduction

Multicore processors make parallelism increasingly attractive, but shared-state parallel programs are difficult to reason about. The programmer wants the performance benefits of running independent computations at the same time, while the language designer wants the semantic simplicity of deterministic execution: well-typed programs should not acquire data races, deadlocks, or schedule-dependent results merely because the implementation exploits parallel hardware. A central challenge is therefore to expose parallelism without asking the programmer to reason directly about low-level interleavings.

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The language studied in this paper is a region-polymorphic λ -calculus with mutable references and executable effect summaries. Typability is established using a type-and-effect discipline in the Lucassen–Gifford lineage (Lucassen and Gifford, 1988). Region effects and non-interference checks for deterministic parallelism are not, by themselves, the novelty claimed here; DPJ, for example, uses modular compile-time type-and-effect checking to guarantee deterministic semantics (Bocchino et al., 2009). The question we ask is more specific: can the programmer provide a checked, executable summary that is more precise than a coarse static region effect, but less intrusive than fully logging or validating every dynamic access?

Throughout the paper, we call the source-level executable effect-summary terms *surface-effect terms*; when the context is clear, we use *surface effects* as shorthand for this term language. Evaluating a surface-effect term yields a *computed effect*, which is the run-time summary value available to later analyses or runtimes. The contribution is not merely that effects appear in types, but that a computation can be paired with another well-typed source term that computes a summary of the computation before the computation itself is run.

Two degenerate surface-effect terms are useful for intuition. ‘Pure’ denotes the empty summary and is appropriate for computations that perform no dynamic memory actions. ‘Top’ is the opposite extreme: it is a sound summary for any computation, but it gives up precision and therefore prevents a useful address-level comparison. Between these extremes, surface-effect terms can compute value-dependent summaries such as concrete reads and writes. This precision is useful when two computations mention the same static region but access different concrete locations within that region.

The central intuition is that one well-typed term is described by *another* well-typed term of the same language. A summary may inspect run-time values and, for example, compute a concrete region/location pair instead of retaining only a region-level read or write. Such a summary may distinguish accesses that a region-only static effect merges. The formal development does not define a global precision lattice relating static effects, computed effects, and traces, nor does it prove that a more precise summary is necessarily more expensive. The soundness claim is instead expressed by explicit approximation relations: static effects soundly cover dynamic actions, and a verified surface-effect term evaluates to a computed effect that soundly covers the trace of its paired computation.

Put simply, the surface-effects approach is a hybrid design point for value-dependent effect prediction. Purely static approaches are modular and perform the conflict check at compile time, but they may be conservative when pointer-rich data structures collapse many concrete locations into the same abstract region. Purely dynamic and optimistic approaches observe concrete accesses directly, and hence can be very precise, but they usually require logging, validation, rollback, or replay. Surface effects place the check in between: a source-level summary is supplied and verified statically, evaluated before the paired computation runs, and then used as a more precise dynamic summary of the computation’s future accesses.

Figure 1 previews this design point symbolically. The notation is introduced fully in Sections 2–4. Here it is only schematic: $s \longrightarrow s'; \eta$ denotes one step of the operational machine together with the trace fragment η emitted by that step, while $s \longrightarrow^* s'; \phi$ denotes a finite execution whose fragments have been accumulated into the structured trace ϕ . Static checking compares region effects ϵ_i . Surface-effect checking first verifies source expressions against surface-effect terms $\hat{\Theta}_i$, then evaluates those terms by the same operational machine to computed effects Θ_i . Dynamic approaches compare traces ϕ_i obtained from the actual executions. The table writes $\otimes$ for the relevant comparison at each level; the formal development later refines the computed-effect comparison into computed-effect disjointness and non-conflict premises.

Static effects are inexpensive and available at compile time, but their region-level approximation may reject independent concrete accesses. Surface effects pay the cost of evaluating executable summaries before running the paired computations, but can recover value-dependent precision. Dynamic tracking observes the concrete accesses themselves, but usually requires logging, validation, rollback, or replay.

Operationally, the ordinary continuation-machine relation $s \longrightarrow s'; \eta$ evaluates expression structure and emits the trace fragment η . A surface-effect term is evaluated by this same machine; there is no separate summary evaluator. Thus the paper’s runtime theorem is an operational theorem about finite small-step executions, not merely a statement about typing rules.

The static semantics uses type-and-effect judgments of the form $\Gamma \vdash e : \tau ! \epsilon$, where Γ is a typing context, e is an expression, τ is a type, and ϵ is a static effect (see Section 5). The static semantics

	$e_1 \quad e_2$	
	Three places to compare effects	
Static region effects	$\Gamma \vdash e_1 : \tau_1 ! \epsilon_1$	$\epsilon_1 \otimes \epsilon_2 \quad \Gamma \vdash e_2 : \tau_2 ! \epsilon_2$
Surface effects	$\Gamma; \Delta; P \vdash e_1 \blacktriangleleft \widehat{\Theta}_1$ $\widehat{\Theta}_1 \longrightarrow^* \Theta_1$ $\phi_1 \sqsubseteq \Theta_1$	$\Theta_1 \otimes \Theta_2 \quad \Gamma; \Delta; P \vdash e_2 \blacktriangleleft \widehat{\Theta}_2$ $\widehat{\Theta}_2 \longrightarrow^* \Theta_2$ $\phi_2 \sqsubseteq \Theta_2$
Dynamic traces	$e_1 \rightsquigarrow \phi_1$	$\phi_1 \otimes \phi_2 \quad e_2 \rightsquigarrow \phi_2$

Figure 1: Surface effects as a middle ground between static region effects and dynamic access tracking.

of surface effects defines the back-triangle relation ($\blacktriangleleft$), which states that a computational expression e is correctly described by a surface-effect term $\widehat{\Theta}$. At run time, that term is evaluated by the same small-step machine as every other expression. When it returns an effect value, that value is the computed effect Θ . If a computed effect is too imprecise, for example because a summary evaluated to $\top$, the result remains sound but conservative.

Static region/effect systems such as DPJ make non-interference a modular compile-time obligation, but region-level summaries may merge concrete accesses that are distinct at run time (Bocchino et al., 2009). Runtime inspector-executor techniques address irregular dependences by discovering conflict-free work before executing it in parallel (Arenaz et al., 2004). Surface effects occupy a related middle position: the access summary is supplied as a typed source-language term, checked against the computation, and evaluated before the described computation is run. Unlike optimistic post-validation, the calculus does not start the described computation and then roll it back after discovering a summary conflict; however, this fact alone does not establish that pre-checking is cheaper in total.

The formal contribution is therefore a sound scheduling interface rather than an asymptotic performance result. At compile time, an inductive correctness relation validates programmer-written computation/summary pairs. At run time, the summary terms evaluate to computed effects that may contain concrete region/location information. The mechanized development connects these stages: summary correctness gives trace approximation, summary-level safety gives a deterministic parallel trace, and deterministic trace replay gives equivalent final heaps. False positives remain possible because summaries are approximations and ‘Top’ is deliberately maximally conservative.

The contributions of this paper are:

- an executable effect-summary mini-language embedded in a region-polymorphic calculus with mutable references;
- an inductive correctness relation for statically verifying a computation against a programmer-written, value-dependent summary term;
- a trace-emitting small-step operational semantics in which executable summaries are ordinary expressions of type eff ;
- a mechanized approximation theorem showing that a verified summary soundly covers the dynamic trace of its computation;
- a mechanized continuation-machine safety layer showing that every finite prefix of a well-typed execution remains type safe and not stuck, plus explicit terminal determinism and terminal correctness theorems for ordinary executions.

The rest of this paper is organized as follows. We start by presenting a motivating example in Section 1. Section 2 distinguishes static effects, dynamic traces, surface-effect terms, and computed effects. The syntax for types and the surface language, including the surface-effect language, is given in Section 3. The static semantics of the type-and-effect system is presented in Section 5, and the operational semantics of the ordinary continuation machine is presented in Section 4. The computed-effect conflict and disjointness are introduced in Section 2 before presenting the correctness theorem for surface-effect terms in Section 6. The small-step finite-prefix safety layer is summarized in Section 7. We conclude in Section 9 after discussion of related work.

1 Motivating Example

The ML-like program in Fig. 2 illustrates the intuition behind surface-effect terms. The function `incr` increments the integer stored in a mutable reference. Its argument has type `ref int at  $\rho$` , and the latent static effect of the body is $\{R(\rho), W(\rho)\}$: the reference is both read and written. A conventional region-effect check therefore treats two calls to `incr` at the same region as potentially conflicting.

This is conservative. A list of references is usually homogeneous in its region parameter, so two variables `x` and `y` drawn from the same list may both have type `ref int at  $\rho$` . A static region effect cannot distinguish whether `x` and `y` are aliases or two different addresses inside the same region. Surface effects are intended to recover that concrete-address information before the described computation is run.

```
fun incr [rho] (r : int ref at rho) : unit =
  r := !r + 1

fun incr_eff [rho] (r : int ref at rho) : eff =
   $\widehat{\text{Read}(r)} \oplus \widehat{\text{Write}(r)}$ 

fun use (x : int ref at 7) : unit =
  incr[7] x (* described by incr_eff[7] x *)
```

Figure 2: Motivating example: a computation paired with an executable surface-effect summary.

The role of `incr_eff` is to compute the summary that the static region effect cannot express. The surface-effect term $\widehat{\text{Read}(r)} \oplus \widehat{\text{Write}(r)}$ is evaluated as an ordinary effect-valued expression. Its result is a computed effect that mentions the concrete region/location pair of the actual argument. Thus, two calls whose arguments are distinct locations in the same region can produce non-conflicting computed effects, while two calls on the same location still conflict.

The point of the example is not that every invocation of `incr` is safe to parallelize. Statically, each call has the region effect $\{R(\rho), W(\rho)\}$, so two calls at the same region appear to conflict. The surface-effect term `incr_eff` refines that coarse description by computing the summary $\widehat{\text{Read}(r)} \oplus \widehat{\text{Write}(r)}$ from the actual argument `r`. The computed effect can mention concrete region/location pairs rather than only the region ρ .

The same example also illustrates the precision spectrum. A pure computation may be summarized by ‘Pure’, the empty surface-effect term. The computation `incr` cannot soundly use ‘Pure’, because it reads and writes its argument. It could, however, always use ‘Top’: this summary is sound but maximally conservative, so any later consumer of the summary would learn no useful address-level information. The useful summary lies between these extremes: $\widehat{\text{Read}(r)} \oplus \widehat{\text{Write}(r)}$ is still sound, but it exposes enough run-time address information to avoid some false conflicts.

Informally, if `incr_eff` correctly describes `incr`, then the static relation (`incr` $\blacktriangleleft$ `incr_eff`) must hold. The precision slogan is

$$\epsilon \sqsupseteq \Theta \sqsupseteq \phi,$$

but this is not a single global order. It abbreviates two typed bridges: a static effect ϵ covers the region erasure of a computed effect Θ , and the computed effect Θ covers the dynamic trace ϕ by the relation $\phi \sqsubseteq \Theta$. In the motivating case, the programmer-written summary is checked once as part of the program, but its value-dependent result is computed at run time by the ordinary machine.

Finally, surface-effect terms used as summaries are required to be read-only. The purpose of `incr_eff` is to inspect the reference value well enough to construct a summary, not to perform the update that `incr` will perform. This restriction is what lets the metatheory separate summary evaluation from the computation it describes: evaluating the summary can affect the scheduling decision, but it cannot itself perform the writes whose safety is being checked.

2 Static Effects, Dynamic Traces, Surface Effects, and Computed Effects

This section fixes the vocabulary that is only previewed in the Introduction. The development uses four related, but deliberately distinct, notions of “effect”: static effects, dynamic traces, surface-effect terms, and computed effects. The distinction is essential. Static effects are compile-time approximations; dynamic traces are observations of execution; surface-effect terms are source syntax; and computed effects are the values obtained by running that syntax.

Regions, addresses, and actions. We write ρ for a region variable, r for a run-time region constant, l for a heap location, and $l@r$ for the concrete address consisting of location l in region r . A static region p may be either a region variable or a region constant. Static actions describe allocation, read, and write at the level of regions, whereas dynamic actions describe concrete events at allocated addresses. The value component of a dynamic action is important for replaying a trace, but is usually irrelevant for conflict checking, so examples often omit it.

Effect Definitions

ρ	$\in$	$RgnVar$	region variables
r	$\in$	$RgnConst$	region constants
l	$\in$	Loc	heap locations
$l@r$	$\in$	$Addr \triangleq Loc \times RgnConst$	concrete heap addresses
p	$::=$	$\rho \mid r$	static regions
α	$::=$	$A(p) \mid R(p) \mid W(p)$	static actions
ϵ	$\in$	$\mathcal{P}(\alpha)$	static effects
ι	$::=$	$Alloc(l@r, v) \mid Read(l@r, v) \mid Write(l@r, v)$	dynamic actions
ϕ	$::=$	$Nil \mid \iota \mid \phi_1 \bullet \phi_2 \mid \phi_1 \parallel \phi_2$	dynamic traces
θ	$::=$	$\widehat{A}(r) \mid \widehat{R}(r) \mid \widehat{W}(r) \mid \widehat{Read}(l@r) \mid \widehat{Write}(l@r)$	computed actions
Θ	$::=$	$\top \mid \{\theta_1, \dots, \theta_n\}$	computed effects

Static effects and dynamic traces. A static effect ϵ is the effect component of a type-and-effect judgment $\Gamma \vdash e : \tau ! \epsilon$. It abstracts over all possible executions of e using region-level actions such as $R(\rho)$ or $W(\rho)$. A dynamic trace ϕ is produced by the operational semantics. It records the concrete allocation, read, and write events emitted by a particular execution. Structured traces compose sequentially and in parallel; this structure is used by the trace-replay and deterministic-parallelism theorems.

Surface-effect terms and computed effects. A surface-effect term $\widehat{\Theta}$ is an expression of the source language whose result type is eff . Evaluating such a term produces a computed effect Θ . Computed effects intentionally mix abstract region actions and concrete address actions: a summary may say only that a computation writes some location in region r , or it may say that the computation writes the particular address $l@r$. This is the mechanism that lets a summary be more precise than a static region effect while still being evaluated before the paired computation runs.

Not every static or dynamic action has the same status as a computed action. Concrete reads and writes can appear in computed effects, because a summary term can evaluate an argument reference and obtain its region/location pair. Concrete allocation is not used as a computed action in the mechanization; allocation is summarized abstractly at the region level. Conversely, dynamic traces contain concrete allocations, reads, and writes because they describe what an execution actually did.

The surface-effect term ‘Top’ evaluates to the distinguished computed effect $\top$. It is a sound but maximally conservative approximation: every dynamic trace is covered by $\top$, but $\top$ conflicts with every computed effect and cannot satisfy the computed-effect disjointness premise. Thus, a surface-effect term may always fall back to ‘Top’ for soundness, but doing so loses all useful disjointness information. By contrast, ‘Pure’ evaluates to the empty computed-effect set and describes computations with no dynamic actions.

The useful slogan from the earlier presentation is $\epsilon \sqsupseteq \Theta \sqsupseteq \phi$: static effects are coarser than computed effects, and computed effects are coarser than the traces they summarize. In this paper

the slogan is only a guide to the reader; formally, the objects live in different syntactic categories and are related by typed bridges rather than by a single global lattice.

Typed Approximation Bridges

The slogan $\epsilon \sqsupseteq \Theta \sqsupseteq \phi$ abbreviates two typed approximations. The left bridge compares a static effect with the *static erasure* of a finite computed effect:

$$\begin{aligned} \llbracket \widehat{A}(r) \rrbracket_\epsilon &= A(r) & \llbracket \widehat{R}(r) \rrbracket_\epsilon &= R(r) \\ \llbracket \widehat{W}(r) \rrbracket_\epsilon &= W(r) & \llbracket \widehat{\text{Read}}(l@r) \rrbracket_\epsilon &= R(r) \\ \llbracket \widehat{\text{Write}}(l@r) \rrbracket_\epsilon &= W(r). \end{aligned}$$

$$\llbracket \{\theta_1, \dots, \theta_n\} \rrbracket_\epsilon = \{\llbracket \theta_1 \rrbracket_\epsilon, \dots, \llbracket \theta_n \rrbracket_\epsilon\}, \quad \epsilon \sqsupseteq \Theta \text{ iff } \Theta \neq \top \text{ and } \llbracket \Theta \rrbracket_\epsilon \subseteq \epsilon.$$

The right bridge compares a computed effect with a structured dynamic trace. It first forgets only the intensional trace structure:

$$\begin{aligned} \llbracket \text{Nil} \rrbracket_{\text{dyn}} &= \emptyset & \llbracket \iota \rrbracket_{\text{dyn}} &= \{\iota\} \\ \llbracket \phi_1 \bullet \phi_2 \rrbracket_{\text{dyn}} &= \llbracket \phi_1 \rrbracket_{\text{dyn}} \cup \llbracket \phi_2 \rrbracket_{\text{dyn}} & \llbracket \phi_1 \parallel \phi_2 \rrbracket_{\text{dyn}} &= \llbracket \phi_1 \rrbracket_{\text{dyn}} \cup \llbracket \phi_2 \rrbracket_{\text{dyn}}. \end{aligned}$$

This dynamic-action view still contains concrete locations and values, so it is related to computed effects by the structural approximation judgment $\phi \sqsubseteq \Theta$, not by subset inclusion. If we additionally erase dynamic actions to ordinary static actions,

$$\begin{aligned} \llbracket \text{Alloc}(l@r, v) \rrbracket_\epsilon &= A(r) & \llbracket \text{Read}(l@r, v) \rrbracket_\epsilon &= R(r) \\ \llbracket \text{Write}(l@r, v) \rrbracket_\epsilon &= W(r) & \llbracket \phi \rrbracket_\epsilon &= \{\llbracket \iota \rrbracket_\epsilon \mid \iota \in \llbracket \phi \rrbracket_{\text{dyn}}\}, \end{aligned}$$

then subset inclusion becomes well typed at the static level:

$$\phi \sqsubseteq \Theta \text{ and } \Theta \neq \top \implies \llbracket \phi \rrbracket_\epsilon \subseteq \llbracket \Theta \rrbracket_\epsilon.$$

Consequently, when both bridges hold for a finite computed effect, the informal three-level comparison can be read as

$$\llbracket \phi \rrbracket_\epsilon \subseteq \llbracket \Theta \rrbracket_\epsilon \subseteq \epsilon.$$

Computed-effect coverage. The right-hand bridge, $\Theta \sqsupseteq \phi$, is made precise by the structural trace-to-computed-effect judgment $\phi \sqsubseteq \Theta$. This judgment does not evaluate a trace and does not turn an execution trace into a computed effect. Instead, it checks whether a computed effect covers each dynamic action that can occur in the structured trace. The action case delegates to an action-in-summary relation $\iota \preceq_\Theta \Theta$.

Action Coverage

Action-in-summary coverage

$$\iota \preceq_\Theta \Theta$$

ACT-TOP $\frac{}{\iota \preceq_\Theta \top}$	ACT-ALLOC-ABS $\frac{\widehat{A}(r) \in \vartheta}{\text{Alloc}(l@r, v) \preceq_\Theta \vartheta}$	ACT-READ-CONC $\frac{\widehat{\text{Read}}(l@r) \in \vartheta}{\text{Read}(l@r, v) \preceq_\Theta \vartheta}$	ACT-READ-ABS $\frac{\widehat{R}(r) \in \vartheta}{\text{Read}(l@r, v) \preceq_\Theta \vartheta}$	ACT-WRITE-CONC $\frac{\widehat{\text{Write}}(l@r) \in \vartheta}{\text{Write}(l@r, v) \preceq_\Theta \vartheta}$
ACT-WRITE-ABS $\frac{\widehat{W}(r) \in \vartheta}{\text{Write}(l@r, v) \preceq_\Theta \vartheta}$				

Trace Coverage

Structural trace-to-computed-effect coverage

$$\phi \sqsubseteq \Theta$$

TRACE-NIL $\frac{}{\text{Nil} \sqsubseteq \Theta}$	TRACE-ELEM $\frac{\iota \preceq_\Theta \Theta}{\iota \sqsubseteq \Theta}$	TRACE-SEQ $\frac{\phi_1 \sqsubseteq \Theta \quad \phi_2 \sqsubseteq \Theta}{\phi_1 \bullet \phi_2 \sqsubseteq \Theta}$	TRACE-PAR $\frac{\phi_1 \sqsubseteq \Theta \quad \phi_2 \sqsubseteq \Theta}{\phi_1 \parallel \phi_2 \sqsubseteq \Theta}$
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For finite computed effects, the structural judgment has the expected action-set characterization:

$$\phi \sqsubseteq \Theta \iff \forall \iota \in |\phi|_{\text{dyn}}. \iota \preceq_{\Theta} \Theta.$$

This equivalence explains the role of the two trace views above. The dynamic view $|\phi|_{\text{dyn}}$ keeps concrete locations and values and therefore compares with computed effects through $\sqsubseteq$, not through raw subset inclusion. Only after static erasure do ordinary subset comparisons become well typed: if $\phi \sqsubseteq \Theta$ and $\Theta \neq \top$, then $|\phi|_{\epsilon} \subseteq |\Theta|_{\epsilon}$.

The case $\Theta = \top$ is the conservative escape hatch for this right-hand bridge: it covers every dynamic trace, but it is not a finite computed-action set whose erasure recovers useful static precision. The mechanization defines $\top$, the empty computed effect, and union of computed effects, but it does not require a single lattice over ϵ , Θ , and ϕ .

The left-hand bridge $\epsilon \sqsubseteq \Theta$ therefore uses the erasure morphism $[\cdot]_{\epsilon}$ to forget the extra precision of computed effects. It erases both the abstract/concrete distinction and concrete locations: for example, $\widehat{\text{Read}}(l@r)$ and $\widehat{\text{R}}(r)$ both collapse to the static region action $\text{R}(r)$.

The right-hand bridge keeps the concrete actions in the trace and is expressed by $\phi \sqsubseteq \Theta$. This is the bridge established by surface-effect correctness: evaluating a verified surface-effect term produces a computed effect that covers the trace of the paired computation. A later client can combine this bridge with computed-effect disjointness and non-conflict when it wants to reason about separation of two traces.

Computed-effect conflict and disjointness. The computed-effect guard is stated entirely over computed effects. It requires the universal disjointness predicate $\text{Disjoint}(\Theta_1, \Theta_2)$ and the absence of the existential conflict predicate $\Theta_1 \otimes \Theta_2$. We write $\theta_1 \boxtimes \theta_2$ for element-wise conflict between computed actions, using the square-cross symbol $\boxtimes$, and $\Theta_1 \otimes \Theta_2$ for conflict between computed effects.

Existence of conflicts ($\boxtimes$) in θ

$$\frac{\boxed{\theta_1 \boxtimes \theta_2} \quad \text{C-READ-WRITE} \quad \theta = \widehat{\text{Read}}(l@r) \vee \theta = \widehat{\text{Write}}(l@r) \vee \theta = \widehat{\text{R}}(r) \vee \theta = \widehat{\text{W}}(r)}{\widehat{\text{W}}(r) \boxtimes \theta \wedge \widehat{\text{Write}}(l@r) \boxtimes \theta}$$

Existence of conflicts ($\otimes$) in Θ

$$\frac{\boxed{\Theta_1 \otimes \Theta_2} \quad \text{C-EXT-THETA} \quad \frac{\theta_1 \in \Theta_1 \quad \theta_2 \in \Theta_2}{\theta_1 \boxtimes \theta_2}}{\Theta_1 \otimes \Theta_2} \quad \text{C-TOP} \quad \frac{}{\Theta \otimes \top}$$

3 Syntax

The surface language is a region-polymorphic, call-by-value simply typed λ -calculus with mutable references and recursive functions. The distinctive syntactic form is the recursive abstraction $\mu f. \lambda x. \{e_b \mid e_s\}$, which packages two source expressions: the computational body e_b and the surface-effect body e_s . Computational application, written $\text{capp } e_f e_a$, runs e_b ; effect application, written $\text{eapp } e_f e_a$, runs e_s and must produce a value of type eff .

The grammar below presents the surface language in a compact, framed form. It uses the current terminology. A surface-effect term is not a separate run-time language: it is source syntax of type eff . The meta-variable $\widehat{\Theta}$ only isolates the expressions normally used as executable summaries. In the mechanized syntax, Pure is represented by the empty effect expression, while Top is the conservative effect expression.

$$\begin{aligned}
\text{Regions } p &::= \rho \mid r. \\
\text{Types } \tau &::= \text{nat} \mid \text{bool} \mid \text{unit} \mid \text{eff} \\
&\quad \mid \text{ref } p \tau \mid \tau_x \rightarrow \{\tau_c ! \epsilon_c \mid \text{eff} ! \epsilon_s\} \mid \forall \rho. \tau ! \epsilon. \\
\text{Expressions } e &::= x \mid n \mid b \mid \mu f. \lambda x. \{e_b \mid e_s\} \mid \Lambda \rho. e \\
&\quad \mid \text{capp } ee \mid \text{eapp } ee \mid e[p] \mid \text{cond } (e, e, e) \\
&\quad \mid \text{alloc } pe \mid \text{get } pe \mid \text{set } pee \\
&\quad \mid e + e \mid e - e \mid e * e \mid e = e \\
&\quad \mid \hat{\Theta}. \\
\text{Surface-effect terms } \hat{\Theta} &::= \text{Pure} \mid \hat{A}(p) \mid \hat{R}(p) \mid \hat{W}(p) \\
&\quad \mid \widehat{\text{Read}}(e) \mid \widehat{\text{Write}}(e) \mid \hat{\Theta}_1 \oplus \hat{\Theta}_2 \mid \text{Top}.
\end{aligned}$$

Types follow the region-polymorphic simply typed calculus used throughout the formalization, with two additions made explicit here. The base types `nat`, `bool`, and `unit` classify numbers, booleans, and the result of destructive updates; `eff` classifies computed-effect values Θ . A reference type `ref` $p \tau$ describes a mutable cell of contents type τ allocated in region p , where p may be a region variable or a concrete region. Function types record two latent effects: in $\tau_x \rightarrow \{\tau_c ! \epsilon_c \mid \text{eff} ! \epsilon_s\}$, ϵ_c is the static effect of the computational body and ϵ_s is the static effect of the executable summary body. Region-polymorphic types $\forall \rho. \tau ! \epsilon$ abstract over regions and record the latent effect of the body under that abstraction.

Expressions include variables, numeric and Boolean constants, recursive computation/summary abstractions, region abstractions, computational applications, surface-effect applications, region applications, conditionals, arithmetic and equality, heap operations, and surface-effect terms. In $\mu f. \lambda x. \{e_b \mid e_s\}$, the argument variable x and recursive variable f are available in both e_b and e_s . The type system later requires the summary body to be read-only when it is used as the executable summary for a computation.

Returning to Fig. 2, the function `incr` has a computation body that reads and writes its argument, while `incr_eff` is the corresponding surface-effect body. Informally, the static correctness relation must establish $\Gamma; \Delta; P \vdash \text{incr} \blacktriangleleft \text{incr_eff}$. The type of the computation records the coarse static region effect $\{R(\rho), W(\rho)\}$, whereas evaluating the surface-effect body on a particular argument can produce a computed effect mentioning the concrete region/location pair of that argument. The next section gives the small-step semantics that performs this evaluation.

4 Operational Semantics

The operational semantics has one primitive execution relation: the continuation-machine step $s \longrightarrow s'; \eta$. The other judgments below are derived or proof-facing relations built around that step. They are kept distinct because they play different roles in the metatheory.

Layer	Judgment	Role
machine step	$s \longrightarrow s'; \eta$	one continuation-machine step emitting one trace fragment
finite execution	$s \longrightarrow^* s'; \phi$	accumulated machine trace
trace replay	$(\phi, H) \Rightarrow^* (\text{Nil}, H')$	replay of a produced trace over a heap

The notation is intentionally separated. Machine execution *produces* trace fragments; the finite-run relation concatenates them into a structured trace; replay later *consumes* an already produced trace. A one-step transition emits only a fragment η : administrative steps emit `Nil`, while heap operations emit singleton fragments such as `Alloc`($l@r, v$), `Read`($l@r, v$), and `Write`($l@r, v$). The trace of a longer execution is not stored in the state or continuation.

Operational Domains

$E \in Env \triangleq Variables \rightarrow Values$	(variable assignment)
$P \in Rgn \triangleq RgnVars \rightarrow RgnConst$	(regions in scope)
$H \in Heap \triangleq Addr \rightarrow Values$	(heap assignment)
$\langle E, P, e \rangle \in Closures \triangleq Env \times Rgn \times Expressions$	(function closures)
$v \in Values \triangleq \mathbb{N} + \mathbb{B} + Addr + Closures + \Theta + ()$	(dynamic values)
$s ::= \langle H, E, P, e, k \rangle \mid \langle H, v, k \rangle \mid \text{done}(H, v)$	(machine states)
$a ::= \text{Alloc}(l@r, v) \mid \text{Read}(l@r, v) \mid \text{Write}(l@r, v)$	(dynamic actions)
$\eta ::= \text{Nil} \mid a$	(one-step trace fragments)
$\phi ::= \text{Nil} \mid a \mid \phi \bullet \phi \mid \phi \parallel \phi$	(structured traces)

Finite Machine Executions

Finite execution

Each finite execution accumulates one-step fragments by induction.

$$s \longrightarrow^* s'; \phi$$

$$\frac{\text{TR-REFL}}{s \longrightarrow^* s; \text{Nil}} \qquad \frac{\text{TR-STEP} \quad s \longrightarrow s_1; \eta \quad s_1 \longrightarrow^* s_2; \phi}{s \longrightarrow^* s_2; \eta \bullet \phi}$$

Here $\bullet$ is structured trace sequencing and Nil is its identity. Ordinary expression evaluation is the special case

$$\text{init}(H, E, P, e) \longrightarrow^* \text{done}(H', v); \phi.$$

A surface-effect term is just an ordinary expression of type eff ; there is no separate summary evaluator.

Trace Replay

Replay

$$(\phi, H) \Rightarrow^* (\text{Nil}, H')$$

Replay interprets an already produced structured trace as a heap transformer. The following selected rules fix the intended direction: execution produces traces, while replay consumes them.

$$\begin{array}{lll} \text{R-NIL} & \text{R-ALLOC} & \text{R-READ} \\ \frac{}{(\text{Nil}, H) \Rightarrow^* (\text{Nil}, H)} & \frac{}{(\text{Alloc}(l@r, v), H) \Rightarrow^* (\text{Nil}, H[(r, l) \mapsto v])} & \frac{H(r, l) = v}{(\text{Read}(l@r, v), H) \Rightarrow^* (\text{Nil}, H)} \\ \\ \text{R-WRITE} & \text{R-SEQ} & \\ \frac{H(r, l) \neq \perp}{(\text{Write}(l@r, v), H) \Rightarrow^* (\text{Nil}, H[(r, l) \mapsto v])} & \frac{(\phi_1, H) \Rightarrow^* (\text{Nil}, H_1) \quad (\phi_2, H_1) \Rightarrow^* (\text{Nil}, H_2)}{(\phi_1 \bullet \phi_2, H) \Rightarrow^* (\text{Nil}, H_2)} & \end{array}$$

The replay relation for parallel traces allows the two subtraces to be consumed by an interleaving of their replay steps; the determinism theorem later shows that, under the computed-effect guard, terminating replays agree on the final heap.

The initial state for expression e is $\text{init}(H, E, P, e) = \langle H, E, P, e, \text{doneK} \rangle$; terminal states $\text{done}(H, v)$ have no outgoing machine step. Continuations are bookkeeping devices for the call-by-value evaluation order. A frame records the expression still to be evaluated, the environment and region map needed to resume evaluation, and the continuation to resume afterward. Application frames first remember the unevaluated argument while the function position is evaluated, then remember the closure while the argument is evaluated, and finally enter either the computational body or the summary body with the recursive name and argument bound. Heap-operation frames remember the pending operation and the region environment while a location or value subexpression is evaluated. The trace emitted while evaluating functions, arguments, or summaries is the machine trace of that evaluation. A computed effect is the value returned by evaluating the corresponding summary expression.

One machine step

$$s \longrightarrow s'; \eta$$

S-CONST

$$\frac{}{\langle H, E, P, \text{const } n, k \rangle \longrightarrow \langle H, n, k \rangle; \text{Nil}}$$

S-VAR

$$\frac{E(x) = v}{\langle H, E, P, x, k \rangle \longrightarrow \langle H, v, k \rangle; \text{Nil}}$$

S-DONE

$$\frac{}{\langle H, v, \text{doneK} \rangle \longrightarrow \text{done}(H, v); \text{Nil}}$$

S-CLOS

$$\frac{\varphi = \mu f. \lambda x. \{e_b \mid e_s\} \vee \varphi = \Lambda \rho. e}{\langle H, E, P, \varphi, k \rangle \longrightarrow \langle H, \langle E, P, \varphi \rangle, k \rangle; \text{Nil}}$$

S-CAPP-FUN

$$\frac{}{\langle H, E, P, \text{capp } e_f e_a, k \rangle \longrightarrow \langle H, E, P, e_f, K_f^\mu(e_a, E, P, k) \rangle; \text{Nil}}$$

S-CAPP-ARG

$$\frac{v_f = \langle E', P', \mu f. \lambda x. \{e_b \mid e_s\} \rangle}{\langle H, v_f, K_f^\mu(e_a, E, P, k) \rangle \longrightarrow \langle H, E, P, e_a, K_a^\mu(E', P', f, x, e_b, e_s, k) \rangle; \text{Nil}}$$

S-CAPP-BODY

$$\frac{E'' = E'[f \mapsto v_f][x \mapsto v_a] \quad v_f = \langle E', P', \mu f. \lambda x. \{e_b \mid e_s\} \rangle}{\langle H, v_a, K_a^\mu(E', P', f, x, e_b, e_s, k) \rangle \longrightarrow \langle H, E'', P', e_b, k \rangle; \text{Nil}}$$

S-EAPP-FUN

$$\frac{}{\langle H, E, P, \text{eapp } e_f e_a, k \rangle \longrightarrow \langle H, E, P, e_f, K_f^e(e_a, E, P, k) \rangle; \text{Nil}}$$

S-EAPP-ARG

$$\frac{v_f = \langle E', P', \mu f. \lambda x. \{e_b \mid e_s\} \rangle}{\langle H, v_f, K_f^e(e_a, E, P, k) \rangle \longrightarrow \langle H, E, P, e_a, K_a^e(E', P', f, x, e_b, e_s, k) \rangle; \text{Nil}}$$

S-EAPP-BODY

$$\frac{E'' = E'[f \mapsto v_f][x \mapsto v_a] \quad v_f = \langle E', P', \mu f. \lambda x. \{e_b \mid e_s\} \rangle}{\langle H, v_a, K_a^e(E', P', f, x, e_b, e_s, k) \rangle \longrightarrow \langle H, E'', P', e_s, k \rangle; \text{Nil}}$$

S-ALLOC

$$\frac{P(p) = r \quad \text{fresh}(H, r) = l}{\langle H, v, K^{ref}(p, P, k) \rangle \longrightarrow \langle H[(r, l) \mapsto v], l @ r, k \rangle; \text{Alloc}(l @ r, v)}$$

S-GET

$$\frac{P(p) = r \quad H(r, l) = v}{\langle H, l @ r, K^{get}(p, P, k) \rangle \longrightarrow \langle H, v, k \rangle; \text{Read}(l @ r, v)}$$

S-SET

$$\frac{P(p) = r \quad H(r, l) \neq \perp}{\langle H, v, K^{set}(p, l, P, k) \rangle \longrightarrow \langle H[(r, l) \mapsto v], (), k \rangle; \text{Write}(l @ r, v)}$$

Effect-valued expression steps. Surface-effect terms are ordinary expressions whose values are computed effects. Abstract summaries return region-level actions without touching the heap; concrete summaries first evaluate their argument and then return a concrete computed action. These rules are ordinary machine steps and therefore emit Nil.

Effect-Valued Expression Steps

One machine step

$$s \longrightarrow s'; \eta$$

$$\frac{\text{S-ABST-READ} \quad P(p) = r}{\langle H, E, P, \widehat{R}(p), k \rangle \longrightarrow \langle H, R(r), k \rangle; \text{Nil}}$$

$$\frac{\text{S-ABST-WRITE} \quad P(p) = r}{\langle H, E, P, \widehat{W}(p), k \rangle \longrightarrow \langle H, W(r), k \rangle; \text{Nil}}$$

$$\frac{\text{S-ABST-ALLOC} \quad P(p) = r}{\langle H, E, P, \widehat{A}(p), k \rangle \longrightarrow \langle H, A(r), k \rangle; \text{Nil}}$$

$$\frac{\text{S-CONC-READ}}{\langle H, l@r, K^{\text{read}}(k) \rangle \longrightarrow \langle H, \text{Read}(l@r), k \rangle; \text{Nil}}$$

$$\frac{\text{S-CONC-WRITE}}{\langle H, l@r, K^{\text{write}}(k) \rangle \longrightarrow \langle H, \text{Write}(l@r), k \rangle; \text{Nil}}$$

$$\frac{\text{S-CONCAT}}{\langle H, \Theta_2, K^{\text{concat}}(\Theta_1, k) \rangle \longrightarrow \langle H, \Theta_1 \cup \Theta_2, k \rangle; \text{Nil}}$$

$$\frac{\text{S-TOP}}{\langle H, E, P, \text{Top}, k \rangle \longrightarrow \langle H, \top, k \rangle; \text{Nil}}$$

$$\frac{\text{S-EMPTY}}{\langle H, E, P, \text{Pure}, k \rangle \longrightarrow \langle H, \emptyset, k \rangle; \text{Nil}}$$

Effect-summary evaluation is proved read-only from typing: if a well-typed summary execution emits a read-only trace, the final heap is the initial heap. A summary may *denote* an allocation through an abstract allocation action without performing an allocation while the summary itself is evaluated.

5 Static Semantics

The typing rules for the ordinary surface-effect language are given in the present section. A variable context Γ is a total map that records free variables and their types and a region context Δ is a set of region variables in scope. The judgment $\Gamma; \Delta \vdash e : \tau ! \epsilon$ asserts that expression e has type τ under Γ and Δ and the latent (static) effect ϵ . As declared in rule [T-ABST], the correctness between a computational-effect pair is denoted by the binary relation ($\blacktriangleleft$). The rule [T-CAPP] declares the return type of a computational application, **capp**, and the same reasoning is made for **eapp** (whose return type is always **eff**).

Because the mechanization uses logical equality when proving soundness properties about surface-effect terms, types are represented in a locally nameless style (Charguéraud, 2012), so that quantified types are equivalent up to α -conversion. Bound variables are represented with de Bruijn indices and free-variable names are left unchanged. This representation lets substitution be stated directly by type-level β -reduction.

The mechanized predicate **ReadOnly** on static effects is generated by three cases: the empty effect is read-only, a singleton region-read effect is read-only, and unions of read-only effects are read-only. Ordinary static-effect soundness is an auxiliary theorem about evaluation traces: if the static erasure $|\phi|_\epsilon$ of a trace is contained in a read-only static effect, then the trace itself contains only reads. This is separate from the computed-effect precision bridge $\phi \sqsubseteq \Theta$, which deliberately keeps concrete locations until they are erased for static comparison. The corresponding dynamic-trace predicate contains only the empty trace, concrete read actions, and sequential or parallel compositions of read-only traces; allocation and write actions are excluded. Consequently, read-only summary evaluation preserves the heap. This is a semantic property, not a cost claim: the development does not prove that a read-only summary terminates or that evaluating it has constant overhead.

Corollary 1 (Empty static trace erasure implies the empty trace). *For all traces ϕ , if $|\phi|_\epsilon$ is contained in the empty static effect, then $\phi = \text{Nil}$.*

Lemma 1 (Read-only traces of dynamic effects imply no side-effects). *Given that $\text{ReadOnly}(\phi)$, a terminal structured execution*

$$\text{init}(H, E, P, e) \longrightarrow^* \text{done}(H', v); \phi$$

implies that $H \equiv H'$ up to key-value pairs.

Lemma 2 (Read-only static effects imply read-only traces). *If $|\phi|_\epsilon \subseteq \epsilon$ and $\text{ReadOnly}(\epsilon)$ holds for static effects, then $\text{ReadOnly}(\phi)$ holds for the dynamic trace.*

6 Correctness of Surface Effects

The previous section defined the typed bridge $\Theta \sqsupseteq \phi$ between computed effects and dynamic traces. This section explains how that bridge is obtained from source-level verification. The judgment

$$\Gamma; \Delta; P \vdash e \blacktriangleleft \hat{\Theta}$$

relates a computation e to a surface-effect term $\hat{\Theta}$. If e executes with trace ϕ and $\hat{\Theta}$ evaluates to a computed effect Θ , then surface-effect correctness requires

$$\Theta \sqsupseteq \phi,$$

equivalently $\phi \sqsubseteq \Theta$. The rules defining this coverage relation are part of the general effect vocabulary in Section 2; this section proves that verified surface-effect terms establish that relation.

The judgment is intentionally syntax directed but incomplete: there are infinitely many sound summaries for a given expression, and some expressions can always be over-described by ‘Top’. The theorem below therefore states a soundness property for derivable descriptions, not a completeness result for all possible summaries.

The correctness theorem below follows the clean direction of the design. A surface-effect term evaluates to a computed effect, and the computation emits a trace. The theorem proves that the already-computed effect covers the computation trace:

$$\hat{\Theta} \longrightarrow^* \Theta \quad \text{and} \quad \phi \sqsubseteq \Theta.$$

The proof keeps traces and summaries as separate objects. If evaluating a function position, an argument position, or another subexpression may perform memory actions, those actions must be accounted for by the surface-effect term and by the source-level correctness judgment itself; they are not recovered after the fact.

Back-Triangle Rules

Back-triangle rules ($\Gamma; \Delta; P \vdash e \blacktriangleleft \hat{\Theta}$).

In words. The judgment says that the source term $\hat{\Theta}$ is an executable summary for the computation e under the current region map P . The judgment is indexed by the run-time region map P . For a static effect ϵ , write $\epsilon[P]$ for the effect obtained by substituting the region values in P . The following rules give the paper-facing inductive presentation. In particular, pure constants, booleans, variables, abstractions, and region abstractions are described by **Pure**, rather than by an arbitrary summary.

$$\frac{\text{EFF-CONST} \quad \Gamma; \Delta \vdash \text{const } n : \text{nat} ! \emptyset}{\Gamma; \Delta; P \vdash \text{const } n \blacktriangleleft \text{Pure}}$$

$$\frac{\text{EFF-BOOL} \quad \Gamma; \Delta \vdash \text{bool } b : \text{bool} ! \emptyset}{\Gamma; \Delta; P \vdash \text{bool } b \blacktriangleleft \text{Pure}}$$

$$\frac{\text{EFF-VAR} \quad \Gamma; \Delta \vdash \text{var } x : \tau ! \emptyset}{\Gamma; \Delta; P \vdash \text{var } x \blacktriangleleft \text{Pure}}$$

$$\frac{\text{EFF-ABST}}{\Gamma; \Delta; P \vdash \mu f. \lambda x. \{e_b \mid e_e\} \blacktriangleleft \text{Pure}}$$

EFF-RGN-ABST	
$\Gamma; \Delta; P \vdash \lambda \rho. e \blacktriangleleft \text{Pure}$	
EFF-APP	
$\text{ReadOnly}(\epsilon_e[P])$	$\Gamma; \Delta \vdash \text{capp } e_f e_a : \tau_b ! \epsilon_b \quad \Gamma; \Delta \vdash \text{eapp } e_f e_a : \text{eff} ! \epsilon_e$ $\Gamma; \Delta \vdash e_f : \tau_f ! \epsilon_f \quad \Gamma; \Delta \vdash e_a : \tau_a ! \epsilon_a \quad \Gamma; \Delta; P \vdash e_f \blacktriangleleft (\text{eapp } e_f e_a)$ $\Gamma; \Delta; P \vdash e_a \blacktriangleleft (\text{eapp } e_f e_a) \quad \text{ReadOnly}(\epsilon_f[P]) \quad \text{ReadOnly}(\epsilon_a[P])$
$\Gamma; \Delta; P \vdash \text{capp } e_f e_a \blacktriangleleft (\text{eapp } e_f e_a)$	
EFF-RGN-APP	
$\Gamma; \Delta \vdash e_r : \tau ! \epsilon$	$\Gamma; \Delta \vdash \text{Pure} : \text{eff} ! \emptyset \quad \Gamma; \Delta; P \vdash e_r \blacktriangleleft \text{Pure}$ $\Gamma; \Delta; P \vdash e_r[r] \blacktriangleleft \text{Pure}$
EFF-COND	
$\text{ReadOnly}(\epsilon_c[P])$	$\Gamma; \Delta \vdash e_c : \tau_c ! \epsilon_c \quad \Gamma; \Delta \vdash e_t : \tau_t ! \epsilon_t \quad \Gamma; \Delta \vdash e_f : \tau_f ! \epsilon_f$ $\Gamma; \Delta; P \vdash e_c \blacktriangleleft \text{Pure} \quad \Gamma; \Delta; P \vdash e_t \blacktriangleleft e_1 \quad \Gamma; \Delta; P \vdash e_f \blacktriangleleft e_2$
$\Gamma; \Delta; P \vdash \text{cond } e_c e_t e_f \blacktriangleleft \text{cond } e_c e_1 e_2$	
EFF-ALLOC	
$\Gamma; \Delta \vdash e : \tau ! \epsilon$	$\Gamma; \Delta; P \vdash e \blacktriangleleft e_1$
$\Gamma; \Delta; P \vdash \text{alloc } r e \blacktriangleleft e_1 \oplus \widehat{A}(r)$	
EFF-GET-ABS	
$\Gamma; \Delta \vdash e : \tau ! \epsilon$	$\Gamma; \Delta; P \vdash e \blacktriangleleft e_1$
$\Gamma; \Delta; P \vdash \text{get } p e \blacktriangleleft e_1 \oplus \widehat{R}(p)$	
EFF-GET-CONC	
$\Gamma; \Delta \vdash e : \tau ! \epsilon$	$\Gamma; \Delta; P \vdash e \blacktriangleleft \text{Pure}$
$\Gamma; \Delta; P \vdash \text{get } r e \blacktriangleleft \widehat{\text{Read}}(e)$	
EFF-SET-ABS	
$\Gamma; \Delta; P \vdash e_r \blacktriangleleft e_1$	$\Gamma; \Delta; P \vdash e_v \blacktriangleleft e_2 \quad \Gamma; \Delta \vdash e_r : \tau_r ! \epsilon_r \quad \text{ReadOnly}(\epsilon_r[P])$
$\Gamma; \Delta; P \vdash \text{set } p e_r e_v \blacktriangleleft e_1 \oplus (e_2 \oplus \widehat{W}(p))$	
EFF-SET-CONC	
$\Gamma; \Delta; P \vdash e_r \blacktriangleleft e_1$	$\Gamma; \Delta; P \vdash e_v \blacktriangleleft e_2 \quad \Gamma; \Delta \vdash e_r : \tau_r ! \epsilon_r \quad \text{ReadOnly}(\epsilon_r[P])$
$\Gamma; \Delta; P \vdash \text{set } r e_r e_v \blacktriangleleft e_1 \oplus (e_2 \oplus \widehat{\text{Write}}(e_r))$	
EFF-ARITH	
$\Gamma; \Delta \vdash e_2 : \tau_2 ! \epsilon_2$	$\Gamma; \Delta \vdash e_1 : \tau_1 ! \epsilon_1$ $\text{ReadOnly}(\epsilon_1[P]) \quad \Gamma; \Delta; P \vdash e_1 \blacktriangleleft \widehat{\Theta}_1 \quad \Gamma; \Delta; P \vdash e_2 \blacktriangleleft \widehat{\Theta}_2$
$\Gamma; \Delta; P \vdash e_1 \odot e_2 \blacktriangleleft \widehat{\Theta}_1 \oplus \widehat{\Theta}_2$	
EFF-TOP	
$\Gamma; \Delta; P \vdash e \blacktriangleleft \text{Top}$	

Here $\odot \in \{+, -, \times, =\}$ abbreviates the four corresponding arithmetic and equality cases. The concrete read and write rules above use a concrete region value r .

In words. A verified surface-effect term evaluates to a computed effect that covers the entire trace of the paired computation.

Theorem 1 (Correctness of surface-effect terms). *Assume well-formed region, typing, heap-typing, and environment-typing judgments for H , E , P , and e_b . If the computation and summary are related*

by $\Gamma; \Delta; P \vdash e_b \blacktriangleleft e_e$, and

$$\text{init}(H, E, P, e_b) \longrightarrow^* \text{done}(H_b, v_b); \phi_b \quad \text{and} \quad \text{init}(H, E, P, e_e) \longrightarrow^* \text{done}(H_e, \text{eff}(\Theta_e)); \phi_e,$$

with $\text{ReadOnly}(\phi_e)$, then

$$\Theta_e \sqsupseteq \phi_b \quad \text{equivalently} \quad \phi_b \sqsubseteq \Theta_e.$$

The conclusion of the correctness theorem is exactly the right-hand typed bridge. In particular, it does *not* conclude that the computation trace ϕ_b is read-only, nor does it prove the left-hand bridge $\epsilon \sqsupseteq \Theta_e$. The small-step results in Section 7 reuse this approximation story while stating progress, preservation, terminal determinism, and terminal correctness directly over the continuation machine.

Corollary 2 (Read-only, heap-preserving summary evaluation). *For a well-typed computation/effect application pair, static effect soundness and the typing premises of the correctness relation imply $\text{ReadOnly}(\phi_e)$. Therefore summary evaluation preserves the initial heap: $H \equiv H_e$.*

The proof separates two facts: the computation trace is shown to satisfy $\phi_b \sqsubseteq \Theta_e$, while the trace produced by evaluating the effect application is separately shown to be read-only. Thus a computed action such as $A(r)$ may *describe* a future allocation without the evaluation of the summary itself allocating memory.

Trace/summary separation. The previous theorem is the only coverage principle needed for surface-effect correctness: a verified surface-effect term must cover the whole trace of its paired computation. The dynamic-action views $|\phi|_{\text{dyn}}$ and $|\phi|_{\epsilon}$ are used only to state erasure and coverage facts about already produced traces.

In words. If two traces are covered by disjoint and nonconflicting computed effects, then their parallel trace composition is deterministic.

Theorem 2 (Trace-composition soundness). *Let $\phi_1 \sqsubseteq \Theta_1$ and $\phi_2 \sqsubseteq \Theta_2$. If*

$$\text{Disjoint}(\Theta_1, \Theta_2) \quad \text{and} \quad \neg(\Theta_1 \otimes \Theta_2),$$

and $\text{Det}(\phi_1)$ and $\text{Det}(\phi_2)$, then

$$\text{Det}(\phi_1 \parallel \phi_2).$$

The proof first shows that a dynamic read-write or write-write conflict would induce a corresponding computed conflict, contradicting the summary-level premise. It separately transports computed-action disjointness to every cross-trace pair of dynamic actions. The result is the deterministic parallel trace judgment above.

Combining the preceding results gives the trace-level proof dependency:

$$\begin{aligned} & \text{verified computation/summary pair} \\ & \quad \xRightarrow{\phi_i \sqsubseteq \Theta_i} \\ & \implies \frac{\text{Disjoint}(\Theta_1, \Theta_2) \wedge \neg(\Theta_1 \otimes \Theta_2)}{\text{Det}(\phi_1 \parallel \phi_2)} \\ & \quad \xRightarrow{\hspace{1cm}} \\ & \text{equivalent final heaps under terminating trace replay.} \end{aligned}$$

Runtime definitions.

$$\Sigma \triangleq \text{Addr} \multimap \text{Type} \quad (\text{heap type assignment})$$

$$\frac{\text{T-HEAP} \quad \forall l @ r \in H, \Sigma \vdash H(l @ r) : \Sigma(l @ r)}{\star \vdash H : \Sigma}$$

$$\frac{\text{T-HEAP-EMPTY}}{\Sigma \vdash \emptyset : \emptyset}$$

$$\frac{\text{T-HEAP-EXT} \quad \Sigma \vdash E : \Gamma \quad \Sigma \vdash v : \tau}{\Sigma \vdash E, [x \mapsto v] : \Gamma, v : \tau}$$

As previously mentioned, the static relation ($\blacktriangleleft$) is intentionally incomplete: many sound summaries may describe the same computation, while some computations may be related only to a coarse

summary by the supplied inductive rules. Concrete computed actions can refine a region-level distinction and may therefore satisfy computed-effect disjointness even when a region-only comparison would be inconclusive. ‘Top’, by contrast, is a sound approximation of every trace but cannot provide useful disjointness evidence. The current calculus contains no separate static recovery rule for ‘Top’; its role in the mechanized semantics is to conservatively lose precision.

For example, the main purpose of designing a type system where the functional type contains a return type for a computational expression and a return type for an effect expression is to be able to describe any computational application ($\text{capp } e_f e_a$) by its corresponding effect application ($\text{eapp } e_f e_a$). The interesting aspect of assigning the “extended” functional type to the expression e_f is that it may carry a non-empty latent effect for the computational term which can be soundly described by the evaluation of the corresponding “surface” effect term. The returned computed summary of $\text{eapp } e_f e_a$ is produced by the function body’s summary expression. If evaluating e_f or e_a may itself perform memory actions, those actions must be covered by the surface-effect term assigned to the whole computation by the correctness judgment. They are not converted into computed summary components after the fact.

Another example is the expression e_b whose reference type is parametrized by a region p . Two different effect summaries are proven sound: the first is as much precise as the static effect would be (rule [EFF-GET]) and the second is potentially “much” more precise than the static effect (rule [EFF-GET-CONC]). The type of the argument of `get` is the reference type parametrized by a region variable or a region constant. Given the region p , the rule [EFF-GET] applies by chaining the effect summary of the argument with the static surface term $\widehat{R}(p)$. When evaluated, this term yields a value that is equivalent to a de facto static effect. However, we can precisely describe a `get` operation using the surface-effect term $\widehat{\text{Read}}(e)$, which directly takes the argument as input in order to evaluate to a concrete memory address inside some allocated region.

The precision of surface-effect terms can be evaluated by analyzing their dynamic semantics. Consider, for example, the rules [S-Abst-Read] and [S-Conc-Write] in Section 4. The first corresponds to the evaluation of the “static version” of the surface-effect term for the `get` operation and the second corresponds to the evaluation of the “concrete version” of the surface-effect term for the `set` operation. The second rule shows that a concrete address, $l@r$, is being evaluated. Now consider the rules for conflict decision presented in Section 2. When concrete allocated addresses are known, it is possible to decide that the effects `Read` and `Write` do not conflict if the addresses are different even though they belong to the same region, allowing more precision (hence the motivating example presented in Fig. 2). However, if only the region is known, the “static version” of the summary, R and W , does not allow the conflict decision at run-time to be more precise than the static analysis.

7 Small-Step Runtime Safety

The continuation machine supports safety statements over finite executions, not only over completed runs. The ordinary small-step safety invariant is indexed by a store typing Σ . Along a finite trace, allocation may extend Σ ; preservation therefore produces a final store typing Σ' with $\Sigma \subseteq \Sigma'$. We write $s \longrightarrow^* s'; \phi$ for a finite small-step execution whose transition fragments accumulate to ϕ , $\text{init}(H, E, P, e)$ for the initial machine state, and $\text{done}(H, v)$ for a terminal state. The judgment $\Sigma \vdash_{\text{st}} s : \tau$ abbreviates the mechanized typed-state invariant, and $\Sigma \vdash_{\text{tr}} \phi$ states that the trace ϕ is well typed with respect to Σ . The predicate $\text{NotStuck}(s)$ means that s is terminal or has a next small-step transition. The predicate $\text{Shape}_{\Sigma}(v, \tau)$ records that the run-time form of v matches type τ under store typing Σ .

In words. Well-typed ordinary machine states do not get stuck, and ordinary machine steps preserve the typed-state invariant.

Theorem 3 (Progress). *If $\Sigma \vdash_{\text{st}} s : \tau$, then either s is terminal or there exist s' and η such that*

$$s \longrightarrow s'; \eta.$$

Theorem 4 (One-step preservation). *If*

$$\Sigma \vdash_{\text{st}} s : \tau \quad \text{and} \quad s \longrightarrow s'; \eta,$$

then $\Sigma \vdash_{\text{st}} s' : \tau$.

Theorem 5 (Terminal trace determinism). *For any ordinary initial state s , if*

$$s \longrightarrow^* \text{done}(H_1, v_1); \phi_1 \quad \text{and} \quad s \longrightarrow^* \text{done}(H_2, v_2); \phi_2,$$

then

$$\phi_1 = \phi_2 \quad \wedge \quad H_1 = H_2 \quad \wedge \quad v_1 = v_2.$$

Theorem 6 (Terminal surface-effect correctness). *Assume the counted-run correctness invariant holds for every natural bound. If*

$$\Gamma; \Delta \vdash e \blacktriangleleft \hat{\Theta}, \quad \text{init}(H, E, P, e) \longrightarrow^* \text{done}(H', v); \phi,$$

and the summary expression terminates with computed effect Θ ,

$$\text{init}(H, E, P, \hat{\Theta}) \longrightarrow^* \text{done}(H_s, \text{eff}(\Theta)); \phi_s,$$

then the computation trace is covered by the computed effect:

$$\phi \sqsubseteq \Theta.$$

The structured-trace version follows from the raw trace statement by erasing the structured view to its underlying machine trace. The premise about the counted-run correctness invariant is the mechanized induction closure used by the ordinary small-step proof; it is not an operational assumption and it is not evidence extracted from a trace.

8 Discussion and Related Work

The closest static comparator is Deterministic Parallel Java (DPJ) (Bocchino et al., 2009). DPJ uses a modular compile-time type-and-effect system to guarantee deterministic semantics and expresses effects over regions. Surface effects retain static verification but move a value-dependent access summary into an executable source-level term. From the run-time side, classic inspector-executor work analyzes irregular dependences before the executor runs conflict-free work in parallel (Arenaz et al., 2004). The distinctive point of the present formulation is therefore not region effects or pre-execution access prediction in isolation, but the integration of a programmer-written, typed summary term with an approximation theorem and a trace-composition proof.

This positioning also exposes a limitation. Expressive region/effect annotations can be difficult to write manually, and prior work has investigated inference precisely to reduce that burden (Tzannes et al., 2019). The present calculus assumes programmer-written surface-effect summaries and does not give an inference or synthesis algorithm. Annotation effort and the frequency with which summaries collapse to ‘Top’ are empirical questions left open here.

Hybrid verification platforms are also close in spirit (Kawaguchi et al., 2012; Flanagan et al., 2006). They use expressive specifications, such as refinement types, to combine static reasoning with value-sensitive properties. Liquid Types support deterministic fine-grained multithreading by expressing effects in first-order formulas over a decidable arithmetic theory. At a higher-level object-oriented setting, HOOL supports expressive object specifications, including subtyping, refinements, and object invariants. As in our calculus, these specifications are pure: evaluating them does not update mutable data.

The design is hybrid in a precise sense. Verification of the computation/summary relation is static, while the summary value is dynamic: the summary term is evaluated with run-time values and its computed effect can be checked for disjointness and conflict absence by later clients. Summary evaluation is still run-time computation and may itself be input dependent. The continuation-machine metatheory proves progress, preservation, terminal determinism, and terminal surface-effect correctness for ordinary executions. It does not prove termination of summary terms, bound their cost, or compare that cost with dynamic validation mechanisms. A future adequacy theorem could relate terminal machine runs to the archived terminating evaluator, but that bridge is not used as a paper theorem here.

Recent work on hybrid type and effect systems applied to concurrent programming is found in Long et al. (2013). The approach defines special static effects, designated open effects, that prevent object-oriented programs from being rejected when no static upper bound in the form of

effect specification is available. More specifically, open effects are assumed not to conflict with any other effects and are used when the dynamic type of receiver methods cannot be known statically due to dynamically dispatched method invocation. Although open effects are somewhat analogous to the surface-effects approach, in the sense that disjointness information may be unknown until runtime, there is a subtle difference: while open effects are inductively defined and designed for optimistic parallelism, where concretized effects soundly approximate runtime effects, surface-effect terms are syntactically defined in terms of unevaluated expressions and their concretization derives from runtime heap allocation. The correctness of surface-effect terms is always statically known, and the purpose of dynamic checking is to find further possibilities for parallelism. Moreover, surface-effect terms can be used in those scenarios where conflict decision based on static effects would reject parallel programs. Since more precise effect specifications, yet statically verified, are provided directly by the programmer, surface-effect analysis can override static-effect analysis.

Irregular data structures such as trees, queues, and graphs motivate the use of run-time information because region-level analyses may merge accesses that are concrete-address disjoint. The Galois approach (Kulkarni et al., 2007) is a useful contrast because it uses run-time conflict detection in an optimistic execution model. The surface-effects approach instead evaluates a summary before the parallel computation and therefore do not require the calculus to roll back a computation after a summary-level conflict is found. This is a semantic difference, not an established performance comparison.

The present work supplies neither a cost semantics nor an implementation study. Consequently, we make no claim that summary evaluation is constant in the size of the summary, that pre-checking has lower total overhead than access logging, or that useful parallelism is recovered frequently in practice. A future evaluation should measure at least summary-evaluation time, conflict-test time, false-positive rate, the frequency of ‘Top’, recovered parallelism relative to region-only static effects, and programmer annotation effort. These measurements are necessary to determine whether the formal precision mechanism translates into a practical advantage.

9 Conclusions

We have presented a region-polymorphic calculus in which a computation is paired with a statically verified, executable effect-summary term. Evaluating the summary before the computation can expose concrete region/location actions that a region-only static effect does not distinguish. The mechanized correctness theorem proves trace approximation, and the trace-composition lemma makes the bridge explicit: disjoint and non-conflicting computed summaries of deterministic component traces yield a deterministic parallel trace. The trace semantics then establishes equivalent terminating replay heaps. The continuation-machine theorem stack adds ordinary runtime safety: well-typed states make progress, machine steps preserve typing, and two terminal executions from the same initial state have the same trace, heap, and value.

These results are formal, not empirical performance results. The calculus does not include summary inference, a termination argument for summary evaluation, a cost semantics, an implementation evaluation, or a separate adequacy theorem for terminal continuation-machine runs. Establishing the annotation burden and the run-time trade-off between summary precision, pre-checking cost, and recovered parallelism is therefore left to future work.

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A Deterministic Parallelism

Definition 1 (Heap equivalence). *The mechanization represents a heap as a finite map from keys (r, l) to values. We write $H_1 \equiv H_2$ when the two finite maps are extensionally equal, equivalently when every key has the same lookup result in both heaps:*

$$H_1 \equiv H_2 \iff \forall k. H_1(k) = H_2(k).$$

This is equality of key–value maps, not an isomorphism modulo renaming of locations. The relation is reflexive, symmetric, and transitive.

Definition 2 (Deterministic trace). *The predicate $\text{Det}(\phi)$ is inductively generated by the empty trace, single dynamic actions, sequential composition of deterministic traces, and a parallel constructor. The latter requires deterministic component traces and both*

$$\neg \text{ConflictTraces}(\phi_1, \phi_2) \quad \text{and} \quad \text{DisjointTraces}(\phi_1, \phi_2).$$

Thus absence of an explicitly represented read–write or write–write conflict is not the only premise: every pair of dynamic actions drawn from opposite branches must also satisfy dynamic-action disjointness.

Definition 3 (Trace replay). *The relation $\Rightarrow$ executes one dynamic action of a trace against a heap. Allocation and write actions update the heap, reads require the recorded value to be present, and sequential/parallel trace nodes provide the evaluation contexts. We write $\Rightarrow^*$ for its reflexive transitive closure. A terminating replay has the form $(\phi, H) \Rightarrow^* (\text{Nil}, H')$.*

Lemma 3 (Read-only trace properties). *A read-only trace preserves the heap during replay. Two read-only component traces form a deterministic parallel trace. These results follow from the inductive definition of ReadOnly on traces, which contains only the empty trace, read actions, and sequential/parallel compositions of read-only traces.*

Lemma 4 (Scheduler soundness). *For $i \in \{1, 2\}$, let $\phi_i \sqsubseteq \Theta_i$ and $\text{Det}(\phi_i)$. If*

$$\text{Disjoint}(\Theta_1, \Theta_2) \quad \wedge \quad \neg(\Theta_1 \otimes \Theta_2),$$

then $\text{Det}(\phi_1 \parallel \phi_2)$.

The proof uses the soundness relation $\sqsubseteq$ in both directions required by the parallel-trace constructor: a dynamic conflict would imply a computed conflict, and computed-action disjointness implies disjointness of every represented pair of dynamic actions.

Theorem 7 (Confluence of deterministic trace replay). *Let $H_a \equiv H_b$. If a deterministic trace ϕ is replayed from the two equivalent heaps to intermediate configurations (ϕ_1, H_1) and (ϕ_2, H_2) , then there exist a common trace ϕ_3 and heaps $H_3 \equiv H_4$ such that*

$$(\phi_1, H_1) \Rightarrow^* (\phi_3, H_3) \quad \text{and} \quad (\phi_2, H_2) \Rightarrow^* (\phi_3, H_4).$$

In particular, if both replays terminate at Nil, their final heaps are equivalent.

The result does not claim uniqueness of an intermediate trace independently of the stated confluence property.

Theorem 8 (Heap uniqueness for safe parallel summaries). *Let $\phi_1 \sqsubseteq \Theta_1$ and $\phi_2 \sqsubseteq \Theta_2$, with deterministic component traces and*

$$\text{Disjoint}(\Theta_1, \Theta_2) \quad \wedge \quad \neg(\Theta_1 \otimes \Theta_2).$$

Let $H_a \equiv H_b$. If

$$(\phi_1 \parallel \phi_2, H_a) \Rightarrow^* (\text{Nil}, H_1) \quad \text{and} \quad (\phi_1 \parallel \phi_2, H_b) \Rightarrow^* (\text{Nil}, H_2),$$

then $H_1 \equiv H_2$.

The proof applies trace-composition soundness to establish determinism of the parallel trace and then uses the terminating replay theorem.